Accelerating and Generalising the Numerical Simulation of Stimulated Brillouin Scattering Dynamics

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Abstract

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1 Introduction

1.1 Background

1.1.1 What is Stimulated Brillouin Scattering

Stimulated Brillouin Scattering (SBS) is a nonlinear optical effect that occurs due to the interaction between light and sound within a material parenciteeggletonBrillouinScatteringPart2022. When a strong optical (pump) wave, typically from a laser, travels through a medium, it undergoes Brillouin scattering as it interacts with co-propagating acoustic (density) waves generated within the medium through changes in density induced by strong electric fields. These acoustic waves form a moving refractive index grating that scatters part of the light backwards, producing a backward-propagating, frequency-shifted Stokes wave. The interaction between these three waves creates a strong self-reinforcing amplification of the Stokes and acoustic waves, which leads to exponential growth of the Stokes wave as energy is progressively drained from the pump.

The Brillouin scattering effect most commonly occurs due to the acoustic waves being optically excited. The two main physical effects related to this scattering are as follows: first, acoustic waves compress the medium, which alters its optical permittivity via photoelasticity. Second, acoustic waves deform the physical boundaries of the medium, which directly alters the permittivity at its edges and interfaces.

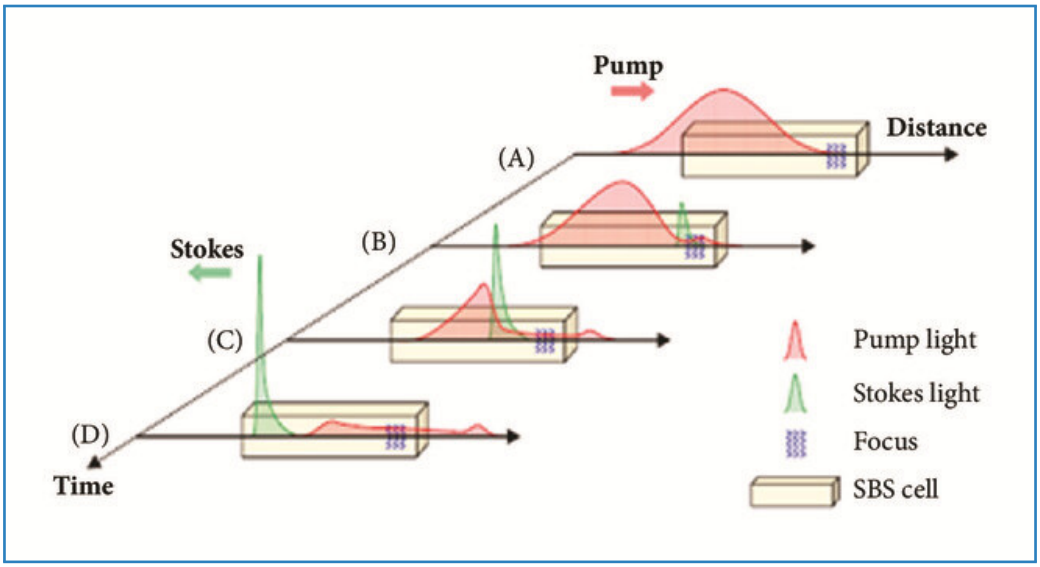


Figure 1.1 | Pump and Stokes wave interaction through time (Bai et al., 2018)

A key result of this scattering effect is that the backward-propagating Stokes wave is frequency-shifted relative to the pump. This is known as the Brillouin Frequency Shift (BFS). It arises because the acoustic waves (acting as an acoustic grating) are moving, so the scattered light experiences a Doppler shift and momentum change. In silica fibre at a wavelength of 1550 nm, this downshift is typically around 11 GHz (Agrawal, 2019). The pump, the acoustic wave and the Stokes wave therefore form a feedback loop, and it is this self-reinforcing amplification that the 'stimulated' in SBS refers to.

1.1.2 Applications of SBS

SBS has many important applications in modern photonics and can act as both a useful tool and a limiting constraint in systems (Bai et al., 2018).

In optical signal processing, SBS can help create narrow-line width fibre lasers and amplifiers, which are essential for high-precision sensing and signal processing in systems where narrow linewidths are important for signal coherence and minimising noise (Agrawal, 2019). Another important application is detecting changes in the environment such as temperature and pressure. This capability makes SBS useful for distributed sensing, such as monitoring the structural health of civil infrastructure. The main approach for such sensing is known as Brillouin Optical Time Domain Analysis.

Brillouin scattering can also have a negative effect on optical systems. Backscattering can occur due to SBS in high-power fibre optic systems, limiting the peak power that can be transmitted without distortion or noise. These effects increase with propagation distance, leading to further distortion. Furthermore, in high-power fibre lasers and amplifiers SBS acts as the dominant effect that caps the output power. This happens once the pump exceeds the SBS threshold and energy is redirected into the backward-propagating Stokes wave rather than the intended output.

Whether trying to suppress or use SBS, accurate simulation of its dynamics is important for understanding the effects within a system. These effects rarely have analytic solutions, thus motivating the need for a generalised simulation framework.

1.1.3 The Main Equations of SBS

SBS dynamics are governed by the three-wave interaction equations, Equation (1.1):

$$\left(\frac{\partial}{\partial T} + V_1\frac{\partial}{\partial Z} + \mu_1\right) a_1 = -\kappa\, a_2 b, \qquad \text{(Pump)} \tag{1.1a}$$

$$\left(\frac{\partial}{\partial T} + V_2\frac{\partial}{\partial Z} + \mu_2\right) a_2 = \kappa\, a_1 b^*, \qquad \text{(Stokes)} \tag{1.1b}$$

$$\left(\frac{\partial}{\partial T} + V_3\frac{\partial}{\partial Z} + \mu_3\right) b = \kappa\, a_1 a_2^*. \qquad \text{(Acoustic)} \tag{1.1c}$$

These equations are a system of coupled, nonlinear partial differential equations, which describe the evolution of the three main wave envelopes. Here a_1, a_2 and b are the slowly varying envelopes of the pump, Stokes and acoustic fields respectively. They are functions of position Z along the waveguide and time T. V_1, V_2 and V_3 are the velocities at which each envelope travels along the waveguide, μ_1, μ_2 and μ_3 are the corresponding linear damping rates, and κ is the coupling coefficient that sets the strength of the interaction between the waves. The asterisk denotes the complex conjugate.

Each equation shares the same structure, where the left-hand side contains a time derivative, a spatial derivative scaled by the respective waves velocity, and a linear damping term. The right-hand side captures the nonlinear coupling between the three waves through the coupling coefficient κ. Energy from the pump is drained by the Stokes and acoustic waves, hence the right-hand side of Equation (1.1a) carries a negative sign. These equations represent a simplified three-wave model that neglects the optical Kerr effect and acoustic noise, to isolate the core dynamics of SBS (Montes, 2008).

Figure 1.2 illustrates the kind of result sought: given the envelopes at the start, the simulation evolves them to give their values at later times.

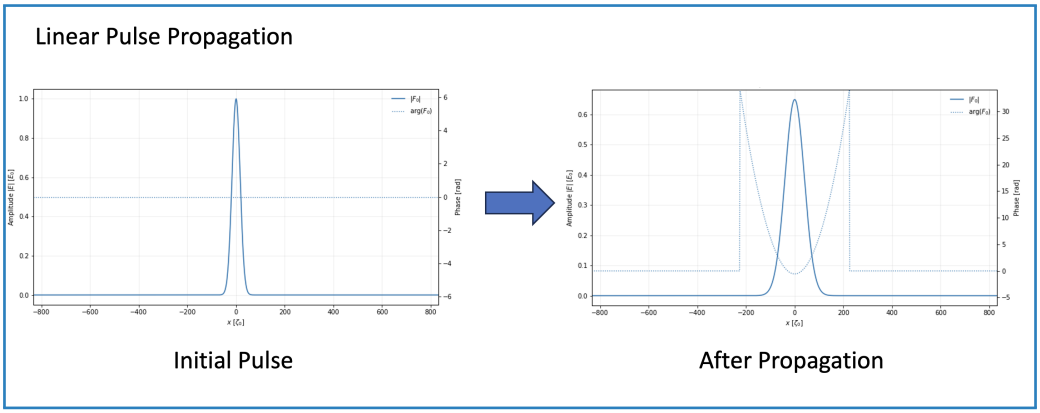


Figure 1.2 | A Gaussian pulse before (left) and after (right) linear propagation. Solid lines show the envelope amplitude and dashed lines show its phase. The simulation takes the initial state as input and computes the evolved state.

Numerical simulation of SBS dynamics can be set up in two ways: following how the waves change in space as time passes, or following how they change in time as they travel along the waveguide. The equations here use the first. They describe how the shape of each wave along the waveguide changes through time. This is the more natural choice for the short structures considered here, where light enters and leaves through the ends, because the state of the whole waveguide can be described at one instant and stepped forward in time (Figure 1.2).

1.2 Why Simulation is Required

Accurately modelling SBS requires solving a set of coupled nonlinear partial differential equations which define the characteristics of the waves. Analytical solutions exist only for simplified cases, such as steady-state operation, where the waves do not change over time (Agrawal, 2019). Once the pump is pulsed or additional effects such as the optical Kerr nonlinearity are included, the nonlinear coupling terms in Equations (1.1a) to (1.1c) prevent the equations from being decoupled, and no general closed-form solution is known. These cases therefore require numerical simulation.

Numerical simulation overcomes these limitations by providing access to the entire evolution of all three wave envelopes, including quantities that are difficult or impossible to measure inside a physical waveguide. It also allows parameters to be varied freely and cheaply, enabling researchers to explore system designs, test suppression strategies or sensing configurations, and compare theory against experiment before committing to costly fabrication or measurement. The closed-form solutions that do exist for specific setups also provide a practical means of testing and validating the simulation's numerical accuracy.

1.3 The Existing C++ Codebase

An existing C++ codebase for numerical simulation of SBS in C++ was created in November 2025 by Professor Michael Steel that provides the basis for which this research builds upon. This existing codebase builds and runs on Linux, Windows, and macOS, and is written in C++ 23 for the core of the simulation, along with some Python scripts for certain testing and setup reasons. The core currently consists of about 14,000 lines of C++ across 97 source files, excluding third-party code, and builds on Eigen for linear algebra, FFTW (MKL on Windows) for Fourier transforms, toml++ for reading the control file, and Catch2 for testing. It simulates how light pulses travel through an optical waveguide and accounts for SBS and Kerr-nonlinear effects. Plots of these waves are output at user defined intervals of the simulation for analysis, and it can simulate many waves, both forward-propagating and backward-propagating at the same time.

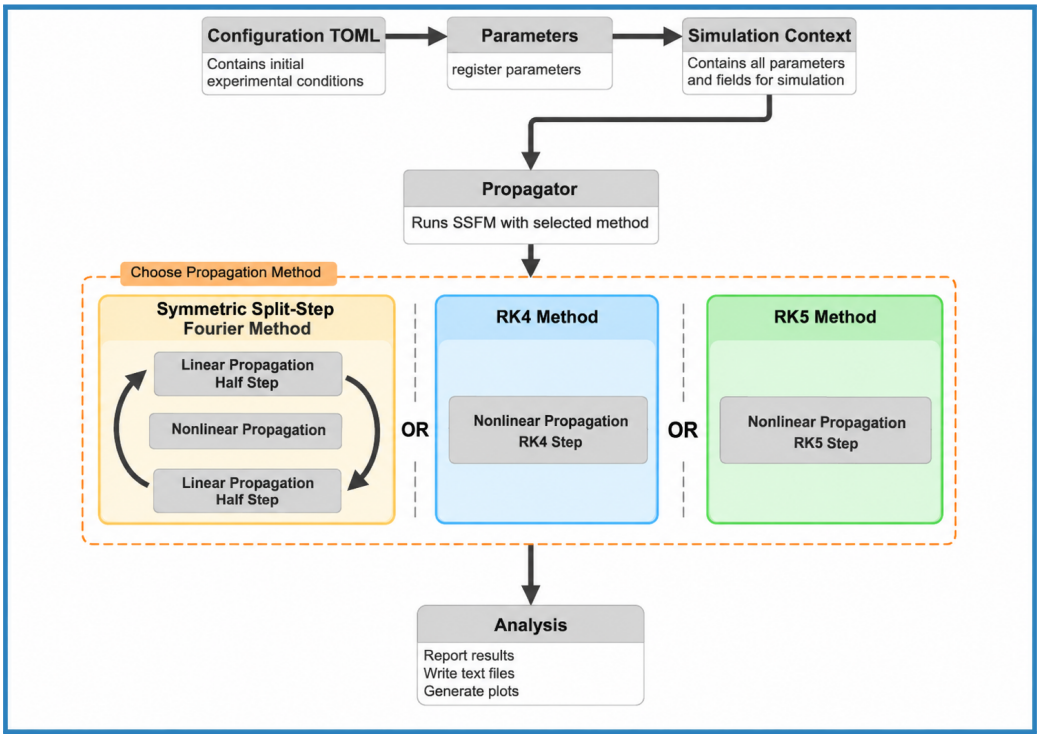


Figure 1.3 | Current codebase architecture schematic.

Figure 1.3 shows a very high-level overview of the codebase architecture. A TOML file configures the specifics of the situation. This file can be used to change a variety of aspects of the simulation, including the active linear and nonlinear effects, forward and backward propagating pulses can be defined to one of the three existing shapes, and adjusting analysis settings. This file is then parsed internally to set up the starting conditions for the simulation in the Parameters and Simulation Context stages. The simulation supports a user choice of propagation methods which define how the effects are applied to the pulses at each time-step; The chosen solver (propagation method) is then used in the Propagation stage and defaults to symmetric split-step if omitted from the configuration file. The chosen propagation is used for the entire simulation for each step. Throughout propagation, the pulse may periodically be printed on a separate thread for analysis, then once the propagation finishes, the final analysis will be done in the Analysis stage.

Most components in the Parameters and Simulation Context stages are default-constructed and then configured by a separate `setup()` call once the parameters they depend on have been parsed. This two-phase initialisation departs from standard C++ practice, under which a constructor should leave an object fully initialised and usable (Stroustrup & Sutter, n.d.). Here, an object is in an unusable state until `setup()` has been called. The pattern is nonetheless common in scientific computing libraries such as PETSc, where objects are created, configured, and then explicitly set up in separate steps (Balay et al., 2025). It allows components to be declared as plain members of the classes that own them and configured later, in the order their dependencies require.

Physical models of SBS involve quantities spanning many orders of magnitude, such as $10^{-27}\,\mathrm{s}^2/\mathrm{m}$, alongside lengths in metres. Entered directly into a computer, these values risk overflow and underflow, and make it hard to tell whether the behaviour of the simulation is correct. These differences in scale are only a side effect of the chosen units: rescaling each quantity by a characteristic value, known as non-dimensionalisation, brings them back to similar sizes. Theorists commonly work this way, while experimentalists generally prefer to work with physical units. The codebase supports both through the `solver.normalise_quantities` option in the TOML file.

1.4 Propagation Methods

The existing SBS simulation has three different propagation methods: symmetric split-step, interaction-picture fourth-order Runge–Kutta (RK4), and interaction-picture Cash–Karp Runge–Kutta (RK5). All three methods solve the same coupled slowly varying envelope equations but differ in how they apply the linear and nonlinear effects. Writing the linear and nonlinear operators as $\mathcal{D}$ and $\mathcal{N}$, advancing the field over a step h is equivalent to applying $e^{h(\mathcal{D}+\mathcal{N})}$. The operators stay the same throughout the simulation, and only the field is updated at each step. On the numerical grid these operators become matrices that do not commute, so $e^{h\mathcal{D}}e^{h\mathcal{N}} \neq e^{h(\mathcal{D}+\mathcal{N})}$, but the two become very close when h is small, which is why the methods take many short steps.

The symmetric split-step method advances the field from z_k to z_{k+1} by applying a half step $(h_k/2)$ of linear effects, followed by a full step (h_k) of nonlinear effects starting from that result, then a second half step of linear effects. Figure 1.4 illustrates the symmetric splitting: the red $\mathcal{D}$ arrows represent linear half-steps, while the blue $\mathcal{N}$ arrows represent full nonlinear steps, and the dashed arrows carry each result to the start of the next operator. Although the nonlinear step is evaluated using RK4, the overall symmetric split-step method is second-order accurate in time.

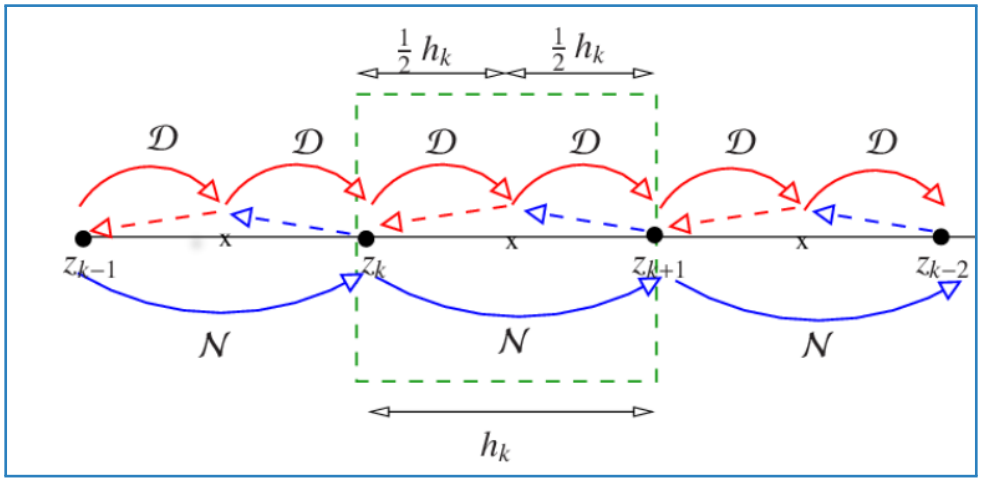


Figure 1.4 | Symmetric split-step propagation, with computational step k shown in the dashed box (Balac & Fernandez, 2014).

The RK4 method is fourth-order accurate in time, so halving the time step is expected to reduce the global temporal error by approximately a factor of sixteen. In comparison, the second-order symmetric split-step method reduces its error by approximately a factor of four. However, RK4 requires more function evaluations per step, meaning its higher convergence order does not necessarily translate to greater computational efficiency.

$$
\begin{aligned}
A^{(i)} &= U^{\mathcal{D}}_{h/2} A_n, \\
k_1 &= U^{\mathcal{D}}_{h/2} f(A_n, z_n), \\
k_2 &= f\left(A^{(i)} + \tfrac{h}{2} k_1,\, z_n + \tfrac{h}{2}\right), \\
k_3 &= f\left(A^{(i)} + \tfrac{h}{2} k_2,\, z_n + \tfrac{h}{2}\right), \\
k_4 &= f\left(U^{\mathcal{D}}_{h/2}\left(A^{(i)} + h k_3\right),\, z_n + h\right), \\
A_{n+1} &= U^{\mathcal{D}}_{h/2}\left[A^{(i)} + \tfrac{h}{6}\left(k_1 + 2k_2 + 2k_3\right)\right] + \tfrac{h}{6} k_4.
\end{aligned}
\tag{1.2}
$$

The fourth-order Runge–Kutta method evaluates the nonlinear derivatives at four intermediate stages, which are then combined using a weighted average and applied to the field, as shown in Equation (1.2). The linear part of the problem is easy: because $\mathcal{D}$ is diagonal in Fourier space, dispersion alone can be solved arbitrarily far ahead in a single step. The nonlinear part cannot be diagonalised and has no such solution except in special cases, so the whole problem cannot be solved in one step. The interaction picture takes advantage of this by instead moving the easy linear part onto the operators, the $U^{\mathcal{D}}_h = e^{h\mathcal{D}}$ terms in Equation (1.2), and leaving only the hard nonlinear part on the field for RK4 to integrate, which reduces the work done per step. Unlike the symmetric split-step method, the RK4 implementation incorporates the linear effects throughout, allowing the interaction between linear and nonlinear effects to be accounted for within each step. This requires four nonlinear derivative evaluations per step, increasing the computational cost per step compared to simpler integration schemes.

Similarly, the RK5 method of propagation is fifth-order accurate in time, so halving the time step reduces the temporal error by approximately a factor of 32. This comes at the cost of additional calculations at each step. Equation (A.1) in Section A.1 gives all six stages of the RK5 method, and Equation (A.2) the fifth-order solution built from them. The Cash–Karp scheme also embeds a fourth-order solution built from the same six stages, and the difference between the two provides a local error estimate that can be used to adapt the step size (Balac & Fernandez, 2014).

1.5 The Research Gap

The focus of this research project is to improve the simulation software in terms of three main measurables: performance, modularity and portability. Program performance is fundamental for timely running of large simulations and reducing the time spent waiting for results. Improving the modularity of the program means it can be easily extended for further research without compromise, for example by adding a new nonlinear effect in one localised part of the code. Portability is important for the program to not just be able to run on any device but to be able to do so efficiently.

Current approaches to simulation software typically focus on creating instruments useful for either theoretical physicists or experimental physicists, rarely addressing the needs of both. Theoretical research focuses on working with idealised or unattainable conditions, whereas experimental research is constrained by real-world limitations. This research project will address the need to create software that meets the requirements of both sides without compromising on either.

When well-developed libraries for simulation do not exist, the common practice in research is to write ad-hoc Python scripts tailored to a specific experimental configuration, use them once, and discard or abandon them for the next task. While Python allows for rapid prototyping, these scripts accumulate technical debt quickly and consequently are rarely used beyond their original purpose. Python's interpreted nature also restricts the performance of any simulations run. This means that each new research task effectively starts from scratch, with researchers reinventing core SBS simulation logic repeatedly rather than building on a stable foundation.

1.6 Research Significance and Physics Outcomes

The significance of this research lies in producing an enhanced simulation codebase for SBS that can serve as a reusable and extensible foundation for future research. Different research requires different environmental configurations, such as wave functions and other nonlinear and linear effects. A flexible and tunable library allows researchers to easily adapt the software to the specific needs of their research, cutting out the time that would be spent rewriting core SBS logic for each specific research task. Well-written and flexible code also means that as new discoveries occur, they can be added to the library without large architectural changes.

From a performance standpoint, the aim of this research project is for all simulations to run faster, leading to less dead time for researchers whilst waiting for results. This outcome is most important for simulations done at a large scale. By writing well-optimised C++ and GPU-accelerated code, not only does the code become more efficient for all simulations, but it can also make it practical to simulate conditions that were previously too large to simulate, potentially revealing unseen effects.

To increase the modularity of the system, on top of writing clean C++ that follows best coding practices, the research will add custom function definitions that can be used to define waves and other spatially dependent fields. This will prevent researchers from being limited to a preset set of options and increases the potential for simulation. Clean coding practices also make the library easy to extend for specific research needs.

"To address portability, all changes will consider the ability of the codebase to be developed and executed across different operating systems (typically macOS or Linux for theorists and Windows for experimentalists)." This will lead to fewer barriers to using and extending the library for all types of physicists.

1.7 Research Objectives

This research is guided by one central objective: to improve the existing SBS modelling codebase into a library that balances performance, modularity for further research, and portability. Success against all three metrics will be evaluated through benchmarking, portability testing, and architectural and structural code analysis against software engineering best practices.

Performance success is defined as a measurable reduction in wall-clock time for the simulation relative to the existing baseline implementation, verified through repeated benchmarking under controlled conditions. Modularity success is defined as the ability to add new physics logic in a localised code fashion, without modification to the code, verified through architectural review and testing. Portability success is defined as the ability to build and run the codebase across macOS, Windows and Linux without any code changes, producing outputs that agree to within floating-point tolerance.

1.8 CPU and GPU Computing

A CPU (central processing unit) is a hardware component responsible for executing computing instructions of a specific instruction set architecture (ISA), performing computations and modifying architectural state. Modern CPUs contain multiple cores, each of which has its own levels of local cache and is capable of independently executing instructions. CPUs are designed for high single-core performance and low execution latency. To achieve this, they use a wide range of techniques such as SIMD instructions, pipelined instruction processing, branch prediction, out-of-order execution, and multi-level cache hierarchies (Stokes, 2006).

A GPU (graphics processing unit), however, is a more specialised piece of hardware originally designed for rendering graphics, which now sees widespread use in machine learning, high-performance computing and other computationally intensive workloads. GPUs are designed for highly parallel computation, containing thousands of arithmetic execution units organised into multiple processing groups. Rather than maximising the computational speed of a single thread, GPUs are designed to maximise computational throughput by executing large numbers of threads simultaneously. This makes a GPU the optimal hardware for large, data-parallel operations where the same computation is applied across many elements (Nickolls & Dally, 2010).

A GPU operates on data held in its own local memory, known as Video Random Access Memory (VRAM), which is physically separate from the system RAM used by the CPU. Data produced on one device and needed by the other must therefore be copied between the two memories over the interconnect that links them, typically a PCIe bus. As illustrated in Figure 1.5, each transfer introduces latency on the order of microseconds, and the available bandwidth is limited to approximately $25\,\mathrm{GB\,s^{-1}}$ for a PCIe 4.0 x16 connection, considerably lower than the bandwidth of the GPU's own memory. The total transfer time therefore depends on both the fixed overhead per transfer and the amount of data being transferred, so workloads that frequently move data between the CPU and GPU can lose much of the benefit of GPU acceleration.

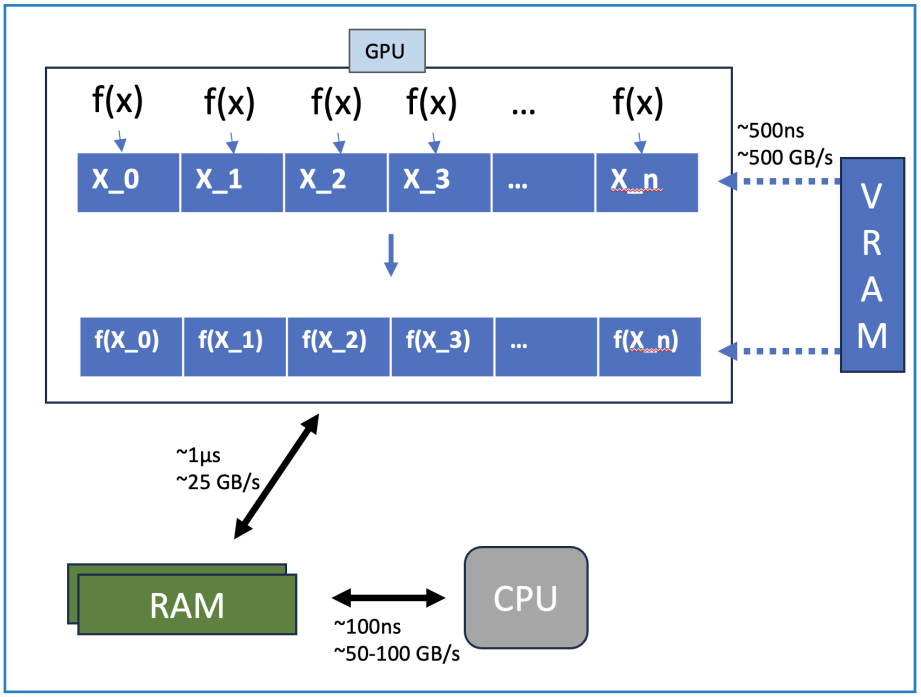


Figure 1.5 | High-level GPU and CPU architecture.

2 Methodology

The work in this project fell into three parts, each covered by a section of this chapter. The first, adding support for custom functions, was also how I became familiar with the codebase. The second added GPU acceleration through LibTorch, and the third built a benchmarking suite to measure the result.

2.1 Custom Function Support

The first stage of improving the codebase was adding support to interpret and apply custom functions to describe initial pulse shapes and spatially dependent couplings. The core goal of the implementation was for researchers to be able to define a function in the input TOML control file as a plain mathematical expression, which is then interpreted by the program and applied to the specified field with minimal overhead. For this we chose to use ExprTk (Expression Toolkit), a third-party C++ library for interpreting mathematical expressions written as strings into functions.

Figure 2.1 shows the previous declaration of a Gaussian-shaped forward-propagating pulse called `launch_f0`. Previously, this pulse was constrained to be in the shape of a Gaussian, super-Gaussian or sech.

```
[launch_fields]
channels = [
  { name = "launch_f0", active = true, shape = "gaussian", ... },
]
```

Figure 2.1 | Old pulse shape specification. Limited to "gaussian", "super-gaussian", or "sech"

These preset wave shapes are still supported and can be chosen without needing to redefine them as custom wave shapes, keeping the TOML file backwards compatible. Custom functions can be added by declaring a function with a name and a definition `f`. As seen in Figure 2.2, the preset names can still be referenced without being defined (the `gaussian` shape in pulse `launch_f1`), but custom shape functions such as `handwrittenGaussian` and `wobblyPulse` can now also be defined and referenced.

```
[spatial_functions]
functions = [
  { name = "handwrittenGaussian", f = "P0 := 2.0; w := 8.0; sqrt(P0) *
  ↪  exp(-x^2 / (2 * w^2))" },
  { name = "wobblyPulse", f = "exp(-x^2 / 50) * (1 + 0.5 * cos(x))" },
]

[launch_fields]
channels = [
  { name = "launch_f0", shape = "handwrittenGaussian", ... },
  { name = "launch_f1", shape = "gaussian", ... },
]
```

Figure 2.2 | Custom defined functions.

Likewise, custom shapes for spatially dependent coupling fields are defined in the same way, then associated with the desired effect in the TOML file, as shown for `kappa` in Figure 2.3.

```
[spatial_functions]
functions = [
  { name = "kapgauss0", f = "w0 := 0.005; kap0 := 10.0; kap0 * sech(x /
  ↪  w0)" },
]

[gratings]
kappa = 10 # constant
# kappa = "kapgauss0" # or a spatial profile
```

Figure 2.3 | Spatially dependent coupling with a custom function.

In the code, both preset and custom shapes are held in a `CustomFunctionStore`, which maps each shape name to a `SpatialFunction` object (Figure 2.4). `SpatialFunction` is an abstract interface with a single `generate` method that returns the shape evaluated across the spatial grid. The presets (Gaussian, super-Gaussian and sech) and the user-defined ExprTk expressions are all implementations of this interface. A launch field therefore requests its shape by name and does not need to know whether it is a preset or a custom function.

```
class SpatialFunction {
    virtual Eigen::VectorXcd generate(const Eigen::VectorXd& x) const = 0;
};

class CustomFunctionStore {
    std::unordered_map<std::string, std::unique_ptr<SpatialFunction>>
    ↪  nameToSpatialFunction;
public:
    Eigen::VectorXcd generate_spatial_function(const Eigen::VectorXd& x,
                                               const std::string& name)
                                               ↪  const;
};
```

Figure 2.4 | Simplified structure of the function store.

Couplings use the same store, but each value in the control file may be either a constant or the name of a function. It is read as a `std::variant<double, std::string>` and resolved with `std::visit`. A number becomes a constant, while a name is looked up in the store and evaluated across the grid. The result is held in a `SpatiallyDependentCoupling`, which wraps a variant of either a single complex scalar or a complex array and overloads the arithmetic operators. The propagation code can therefore apply a coupling without knowing whether it is uniform or spatially varying, and a uniform coupling remains a single scalar at no extra cost.

Custom functions are parsed by ExprTk once at initialisation and evaluated once per grid point to produce the launch and profile arrays. From then on, the propagation uses ordinary arrays, so flexible user input costs only a one-off setup step and nothing per time step, which is negligible compared with the propagation itself.

2.2 GPU Acceleration with LibTorch

The current simulation uses the Eigen library in C++ for linear algebra operations, which runs the entire simulation purely on the CPU. The core calculations of the simulation are the propagation steps, which essentially perform FFTs and element-wise operations over large arrays which hold the pulses. This provides the main question for the performance side of this research: to what extent can the simulation be accelerated with GPU-based computation?

Not all machines have a GPU, and thus moving all computations from Eigen to a GPU-only library would not meet the modularity and portability requirements of the simulation. To add GPU support to the simulation, CPU support must still exist. Additionally, the chosen library should work with as many different GPU vendors as possible, support CPU fallback, and abstract enough away from the hardware that familiarity with GPU computing is not required to contribute to the codebase.

The two strongest libraries for parallelising the program are SYCL and LibTorch. SYCL is a C++-native, hardware-agnostic GPU acceleration library that provides low-level hardware control but is complicated to write. LibTorch is a more abstracted library that gives less control over the hardware, making it far easier to write than SYCL, but it cannot be fine-tuned to get the best possible performance. LibTorch aligned best with the mixed goals of the research, and as such was the chosen library for GPU acceleration.

Adding LibTorch support to the simulation started with how to decide the program should be ran on a GPU. The chosen approach was for opt-in GPU compute via the configuration TOML. This can be set by a simple value as seen in Figure 2.5.

```toml
[general]
compute_backing = "Eigen" # CPU
```

```toml
[general]
compute_backing = "LibTorch" # GPU
```

Figure 2.5 | TOML GPU / CPU computation selection.

This chosen `compute_backing` then determines the way that the data is stored internally in the program. For this, the abstraction of a `FieldData` object was chosen. As shown in Figure 2.6, `FieldData` uses a `std::variant` to hold either an `EigenFieldData` or a `TorchFieldData`, exposing a common public interface for operations on the underlying data. For example, `fft_forward_inplace` applies a forward FFT to each active channel within the first `ncols` columns.

```
class FieldData {
private:
    std::variant<EigenFieldData, TorchFieldData> impl_;

public:
    void fft_forward_inplace(
        const std::vector<int>& active_channels, long ncols, const
        ↪  FFT_Wrapper& fft_wrapper);

    void assign_exp_scaled_col(long col, dcomplex scale, const FieldData&
    ↪  source);

    ...

}
```

Figure 2.6 | Simplified view of a FieldData

This interface decouples the calling code from the underlying storage implementation. Consequently, callers can operate on `FieldData` without needing to know whether the data is backed by Eigen (CPU) or LibTorch (GPU), allowing the same code to work with either hardware.

Simplified representations of `EigenFieldData` and `TorchFieldData` classes can be seen below in Figure 2.7

```
class EigenFieldData {
private:
    Eigen::MatrixXcd data_;

public:
    void fft_forward_inplace(const std::vector<int>& active_channels, long
    ↪  ncols, const FFT_Wrapper& fft_wrapper);

    void assign_exp_scaled_col(long col, dcomplex scale, const
    ↪  EigenFieldData& source);

    ...
}

class TorchFieldData {
private:
    torch::Device device_;
    torch::Tensor data_; // shape [rows, cols], complex double or complex
    ↪  float

public:
    void fft_forward_inplace(const std::vector<int>& active_channels, long
    ↪  ncols, const FFT_Wrapper& fft_wrapper);

    void assign_exp_scaled_col(long col, dcomplex scale, const
    ↪  TorchFieldData& source);
}
```

Figure 2.7 | Simplified view of EigenFieldData & TorchFieldData

Each backing-specific implementation of `FieldData` is required to provide an implementation of every function in the public interface. Calls made through a `FieldData` object are then dispatched to the corresponding function of the underlying type held by `impl_`, using `std::visit`.

This dispatch has minimal runtime overhead compared to a direct function call. The alternatives in `std::variant` are known at compile time, allowing the compiler to generate efficient dispatch code. Also, because the visitor must be valid for every possible alternative of the variant, any missing function implementations result in easily diagnosable compile-time errors.

The usual alternative in C++ is runtime polymorphism, where each backing inherits from a common base class and overrides its virtual functions. The two approaches differ in two main ways. The first is how a call is dispatched. A virtual call looks up the function through the object's vtable at runtime and generally cannot be inlined, whereas `std::visit`, introduced in C++17 alongside `std::variant`, branches on the type the variant currently holds, which the compiler can see and optimise, and checks every operation against every backing at compile time. The second is what is easy to extend. Polymorphism is open to new types but closed to new operations: a new backing is a single new class, but a new operation must be added to the base class and every backing. A `std::variant` is the reverse: a new operation is a single new visitor, but a new backing must be added to the variant and handled by every visitor. For operations over large arrays the cost of the call itself is negligible either way.

The two backings require their own unique implementations for these functions because they don't share an interface. If both LibTorch and Eigen shared an interface, then FieldData could simply be templated on the underlying type of Eigen (`Eigen::MatrixXcd`) or LibTorch (`torch::Tensor`). However by doing so it would also restrict future backing types to those that also share an interface.

The nature of this FieldData also makes it easy to define new operations on the data. Simply declare the function in the interface class `FieldData`, define the backing-specific implementations, and then use the helper function `visit_matched_backings` as seen below in Figure 2.8 to do the `std::visit` dispatch.

```
// Runs `op` on the backings, throwing error for mismatched backing types.
template <typename VariantA, typename Fn, typename... Variants>
decltype(auto) visit_matched_backings(
    const char* opName, Fn&& op, VariantA&& a, Variants&&... rest) {
    if (((a.index() != rest.index()) || ...)) {
        throw std::invalid_argument(std::format("FieldData::{}: backing
        ↪  mismatch", opName));
    }
    return std::visit(
        [&]<typename T>(T& lhs) -> decltype(auto) {
            return op(lhs, std::get<std::decay_t<T>>(rest)...);
        }, a);
}
```

Figure 2.8 | Common `std::visit` helper method

Factoring computation into smaller backing-specific methods requires understanding the trade-off between reusable code and efficient computation. The Eigen library for CPU computation uses expression templates to delay evaluation of computation until the final assignment rather than creating temporary objects. It does this by constructing a lightweight expression object whose type represents the computation at compile time, thus eliminating temporary copies and allocation. Because of this, although creating the `FieldData` interface as a set of simple, basic functions and operator overloads would reduce logic duplication in the code, it would prevent Eigen from using expression templates and regress CPU performance. This is because each function must return a `FieldData`, whose `std::variant` can only hold a fully evaluated `EigenFieldData`, so every call would evaluate its result into a temporary. Hence, the functions in the `FieldData` interface were kept coarse enough that each can be written as a single Eigen expression, such that Eigen operations don't lose their performance. GPU operations through LibTorch don't have the same requirements for their code, as LibTorch evaluates each operation eagerly, launching a GPU kernel per operation regardless of how the code is factored.

When a simulation is run with LibTorch as the selected `compute_backing`, the setup stage determines what hardware to store LibTorch tensors on. This requires determining the type of hardware, as well as specific vendor if GPU. Below Figure 2.9 shows how this flow works. First, the CUDA API checks whether a GPU is available, then the XPU API checks for an Intel GPU; otherwise, the program defaults to the CPU. `resolve_torch_device` uses a lazy initialised static variable constructed from an immediately invoked lambda. This means that the lambda is only called the first time the function runs; subsequent calls return the cached device type.

```
torch::DeviceType resolve_torch_device() {
    static const torch::DeviceType device = [] {
        DeviceType d;
        if (cuda::is_available()) { // NVIDIA GPU
            d = kCUDA;
        } else if (xpu::is_available()) { // intel GPU
            d = kXPU;
        } else {
            d = kCPU; // CPU
        }
        g_logger.info(...);
        return d;
    }();
    return device;
}
```

Figure 2.9 | `resolve_torch_device` simplified function for choosing the most powerful owned device for LibTorch

AMD GPUs are not currently supported by this implementation. LibTorch can use AMD GPUs through ROCm, but ROCm support depends on a compatible Linux environment, an AMD GPU and driver supported by the selected ROCm release, and a matching ROCm-enabled LibTorch build. In theory, PyTorch can expose ROCm devices through its CUDA-compatible API, which would be detected in the existing function. However, this hasn't been tested thoroughly.

When tensors are stored on a GPU, they reside in VRAM rather than system RAM. However, the periodic analysis of the optical pulses during propagation is performed on the CPU, so the pulse data must first be transferred from VRAM to system RAM. Consequently, every analysis step incurs the transfer overhead described in Figure 1.5 in addition to the cost of the analysis itself, and this overhead grows with both the frequency of analysis output and the size of the pulse arrays.

To add a new `compute_backing` type, such as a `SyclFieldData` or a `CudaFieldData`, each function of the public `FieldData` interface would remain unchanged as it already dispatches to the underlying type. Once the new `FieldData` is added to the `std::variant` type of `impl_`, any undefined interface functions of the new type will throw compile time errors. Once all functions are defined, the code will dispatch to the new `compute_backing`. This code makes the simulation codebase very easy to further experiment with different libraries and hardware.

Building of the binary was also designed for cross platform ease of use. Users can enable/disable LibTorch builds via a flag so that users intending of running on CPU only don't require any GPU libraries. This way people without a GPU can still build the program and run simulations without needing to download the GPU libraries.

2.3 Benchmarking Suite

To compare performance across compute backings and simulation configurations, I developed a Python benchmarking suite that runs the simulation over controlled parameter sweeps. The main runner can be specified to vary a specific attribute of the run such as grid size, compute backing, propagation method, or the number of optical channels. For each configuration, the suite repeats the simulation and records its runtime in a CSV file. It captures total simulation time, as well as the time spent in linear and nonlinear parts of each step. A plotting script then interprets the CSV and visualises the results. Figure 2.10 shows the terminal output of a short sweep over three grid sizes with two repeats each. The results generated from this benchmarking suite will be shown in the following chapter on results.

```
$ python testing/benchmarks/run_benchmarks.py --domain-steps 4096,16384,65536 --repeats 2
Building cmake-build-release (Release) ...
...done.
[   1/24] linear_only  Eigen     symsplit     N=4096    rep=0     0.04s  lin=  0.04s nl=  0.00s
[   2/24] linear_only  Eigen     symsplit     N=4096    rep=1     0.04s  lin=  0.04s nl=  0.00s
[   3/24] linear_only  LibTorch  symsplit     N=4096    rep=0     0.07s  lin=  0.07s nl=  0.00s
[   4/24] linear_only  LibTorch  symsplit     N=4096    rep=1     0.07s  lin=  0.07s nl=  0.00s
[   5/24] linear_only  Eigen     symsplit     N=16384   rep=0     0.17s  lin=  0.17s nl=  0.00s
[   6/24] linear_only  Eigen     symsplit     N=16384   rep=1     0.17s  lin=  0.17s nl=  0.00s
[   7/24] linear_only  LibTorch  symsplit     N=16384   rep=0     0.31s  lin=  0.31s nl=  0.00s
[   8/24] linear_only  LibTorch  symsplit     N=16384   rep=1     0.31s  lin=  0.31s nl=  0.00s
[   9/24] linear_only  Eigen     symsplit     N=65536   rep=0     0.85s  lin=  0.84s nl=  0.00s
[  10/24] linear_only  Eigen     symsplit     N=65536   rep=1     0.85s  lin=  0.84s nl=  0.00s
[  11/24] linear_only  LibTorch  symsplit     N=65536   rep=0     1.43s  lin=  1.43s nl=  0.00s
[  12/24] linear_only  LibTorch  symsplit     N=65536   rep=1     1.43s  lin=  1.42s nl=  0.00s
[  13/24] kerr         Eigen     symsplit     N=4096    rep=0     0.07s  lin=  0.04s nl=  0.04s
[  14/24] kerr         Eigen     symsplit     N=4096    rep=1     0.07s  lin=  0.04s nl=  0.04s
[  15/24] kerr         LibTorch  symsplit     N=4096    rep=0     0.26s  lin=  0.07s nl=  0.19s
[  16/24] kerr         LibTorch  symsplit     N=4096    rep=1     0.25s  lin=  0.07s nl=  0.18s
[  17/24] kerr         Eigen     symsplit     N=16384   rep=0     0.31s  lin=  0.17s nl=  0.14s
[  18/24] kerr         Eigen     symsplit     N=16384   rep=1     0.31s  lin=  0.17s nl=  0.14s
[  19/24] kerr         LibTorch  symsplit     N=16384   rep=0     0.89s  lin=  0.31s nl=  0.57s
[  20/24] kerr         LibTorch  symsplit     N=16384   rep=1     0.89s  lin=  0.31s nl=  0.58s
[  21/24] kerr         Eigen     symsplit     N=65536   rep=0     1.45s  lin=  0.84s nl=  0.61s
[  22/24] kerr         Eigen     symsplit     N=65536   rep=1     1.47s  lin=  0.85s nl=  0.62s
[  23/24] kerr         LibTorch  symsplit     N=65536   rep=0     4.52s  lin=  1.46s nl=  3.06s
[  24/24] kerr         LibTorch  symsplit     N=65536   rep=1     4.61s  lin=  1.47s nl=  3.15s

wrote 24 rows to testing/benchmarks/results/results.csv
plot with: python testing/benchmarks/plot_benchmarks.py --csv testing/benchmarks/results/results.csv
```

Figure 2.10 | Terminal output of the benchmarking suite sweeping grid size and compute backing for the linear-only and Kerr control files

3 Results

3.1 Simulation Outputs

Figure 3.1 shows the field amplitude at the start and end of a simulation with a Gaussian input pulse and group-velocity dispersion enabled. The pulse broadens during propagation, while its peak amplitude decreases. This is the expected change for a dispersive pulse and provides a simple example of the field output produced by the simulation.

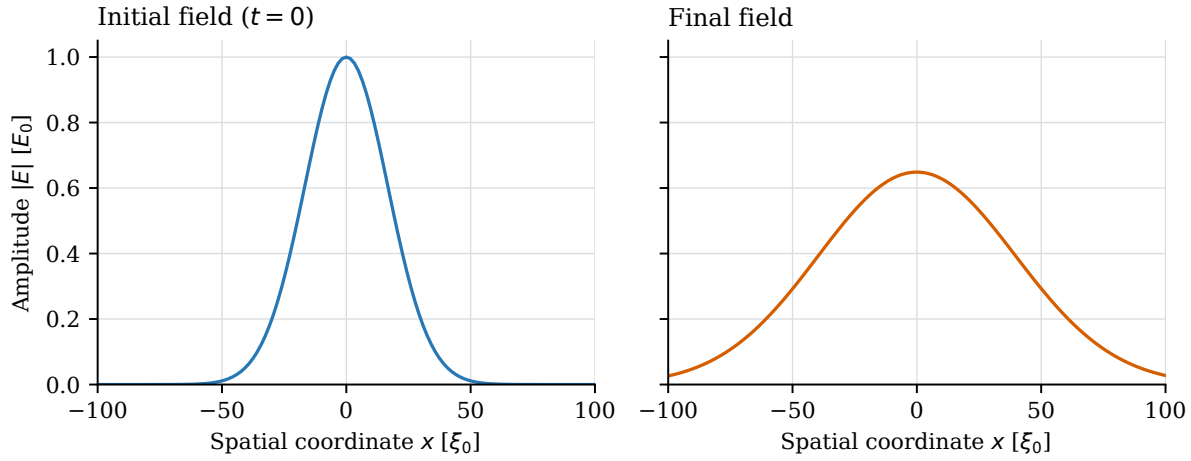


Figure 3.1 | Initial and final field amplitudes for the built-in Gaussian channel in a group-velocity-dispersion simulation. Both panels show the same spatial range, $-100 \leq x/\xi_0 \leq 100$. The final pulse is broader and has a lower peak amplitude.

As discussed earlier, the simulation now allows for custom defined pulse shapes. An example of this can be seen below in Figure 3.2 which shoes the pre and post simulation of these shapes. For this example, the custom amplitude profile is $\exp(-x^2/50)(1+0.5\cos(x))$, the same as seen in Figure 2.2. The two profiles are launched as separate forward channels in the same simulation, illustrating that a custom shape can be used alongside a preset shape.

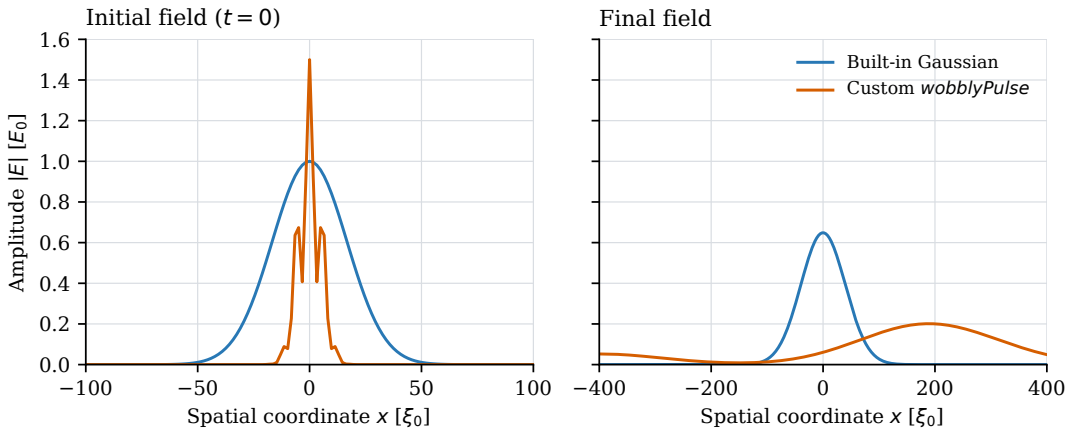


Figure 3.2 | Initial (left) and final (right) field amplitudes for the built-in Gaussian and custom pulse, launched as two forward channels in one simulation. The Gaussian's initial profile is the same as in Figure 3.1. The horizontal ranges differ to show the full profiles before and after propagation.

3.2 Performance Benchmarking

The first performance experiments isolated the linear propagation path by disabling all nonlinear effects. The linear computation was the first part ported to using `FieldData` objects, and hence benchmarking acted as a proof of concept. The experiments compared the Eigen CPU implementation with LibTorch on the GPU while varying the number of spatial grid points and optical channels.

Figure 3.3 shows total simulation time for one optical channel as the grid size increases. At the smallest measured grid size, the GPU was slower than the CPU (a CPU-to-GPU time ratio of 0.5). The GPU became faster as the problem grew: the ratio reached 6.6 at 131,072 grid points and 18.6 at 1,048,576 grid points. Below grid sizes of 2^{15}, the cost of launching a GPU operation becomes the main cost in time, and the actual operations don't become any faster, so the CPU is able to be quicker in those situations. In practice, grid points would be kept above this level for higher precision, and the GPU advantage would be relevant.

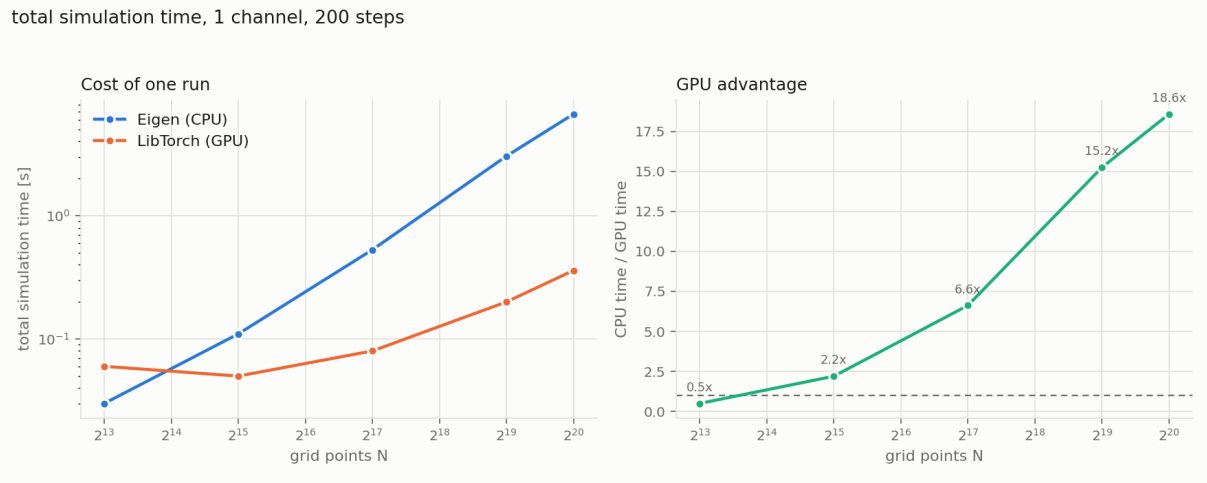


Figure 3.3 | Linear-only simulation time and CPU-to-GPU speedup while sweeping the number of grid points. The simulation uses one optical channel and 200 propagation steps.

The channel-count sweep in Figure 3.4 holds the grid size at 131,072 points and uses 200 propagation steps. The measured GPU speedup ranged from 7.7 for one channel to 19.9 for 128 channels. This increase is consistent with the GPU using its parallel compute resources more effectively as more field data is processed. Together, the two sweeps show that the linear-only GPU implementation benefits from larger grids and more channels, although the size sweep shows that a GPU is not necessarily faster for small problems.

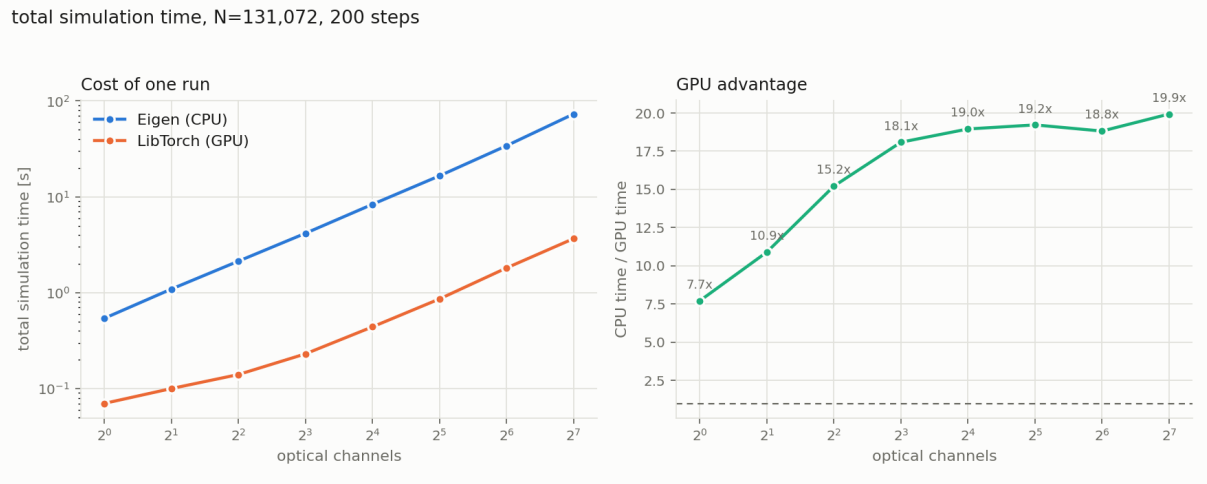


Figure 3.4 | Linear-only simulation time and CPU-to-GPU speedup while sweeping the number of optical channels. The grid contains 131,072 points and the simulation uses 200 propagation steps.

Once the nonlinear effects were able to run across both backings through `FieldData` objects, the next experiment enabled nonlinear effects in addition to the linear propagation terms. This tests whether the performance trend in the linear-only case also holds when the nonlinear calculation is included, and provides comparison between GPU and CPU speed for the program as a whole. As mentioned before, grid sizes below 2^{15} have their timing dominated by kernel launches and that explains why they are seen as unusually slow in the right graph of Figure 3.5.

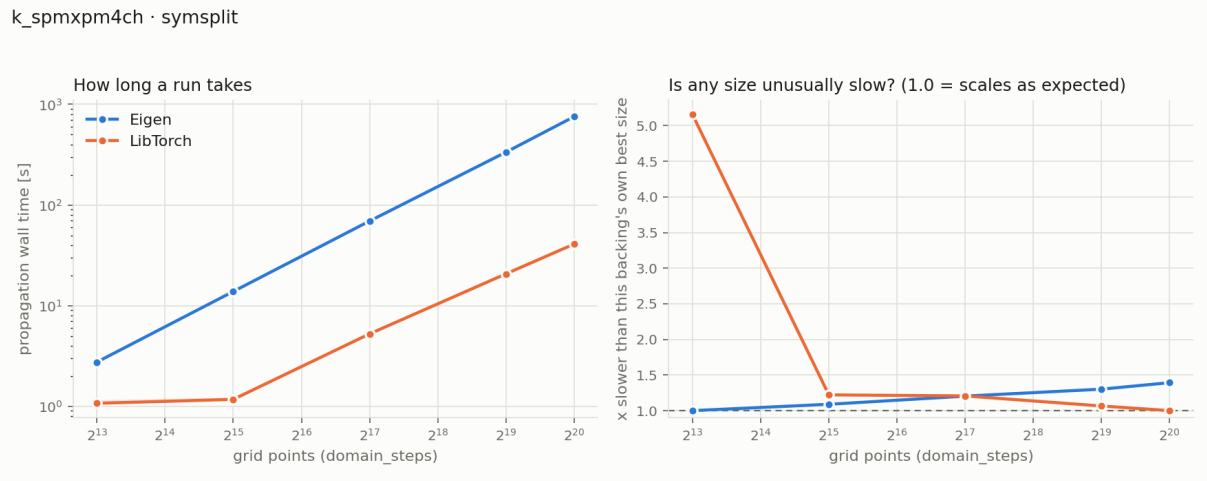


Figure 3.5 | Propagation time for the symmetric split-step method with SPM and XPM enabled across four optical channels. The left panel shows runtime as grid size increases; the right panel compares scaling against each backing's own best result.

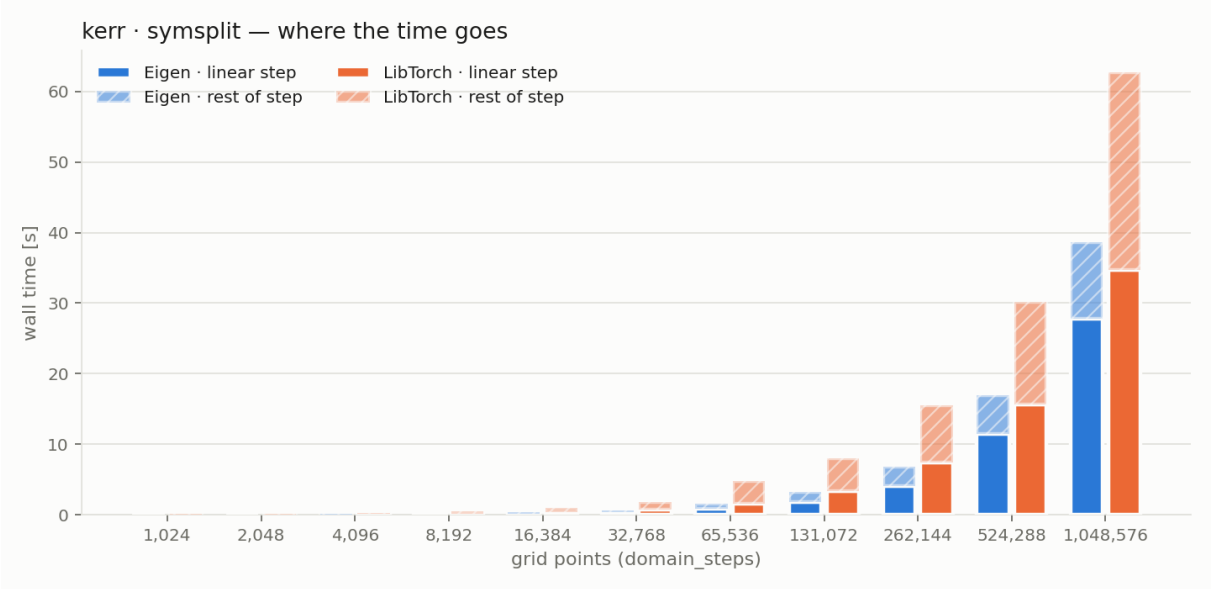


Figure 3.6 | Time spent in the linear step and the remaining work for the Kerr benchmark using symmetric split-step. The remaining-work segment includes the nonlinear calculation and time outside the measured linear step.

3.2.1 Which Parts of the Simulation Take the Most Time?

The timing breakdown in Figure 3.6 compares the linear step with the remaining work in the simulation. The benchmarks show that the linear part receives a greater performance improvement from GPU execution than the nonlinear part. This difference is related to how each part processes the field data (`svea`). In one SPM-only step, the nonlinear calculation performs 39 operations and reads or writes `svea` 84 times. The linear calculation performs 6 operations and reads or writes `svea` 15 times. The repeated field-data accesses in the nonlinear calculation create more memory traffic and more intermediate operations, which limits the performance improvement it gets from the GPU. By comparison, the linear calculation consists largely of FFT operations, which are well suited to GPU execution. The operation counts provide a simple explanation for the difference, although the exact performance improvement depends on the hardware and simulation configuration.

3.2.2 Precision and loss effects

The LibTorch results above use double-precision complex values, matching that of Eigen. The FP64 ratio (64 bit floating-point ratio) refers to the throughput ratio of FP64 arithmetic on a GPU compared to FP32 (32 bit floating-point) arithmetic. Typical consumer GPU cards have an FP64 ratio of 1:64, meaning that double-precision operations take 64 times as long to be performed. Common high-end GPUs in HPC can have FP64 ratio as good as 1:2, such as the NVIDIA V100, A100, or H100. Since double-precision arithmetic can be substantially more expensive on some GPUs than single-precision arithmetic, a further experiment compares LibTorch performance using 64-bit and 32-bit floating-point values. The measured runtimes are shown in Table 3.1. Across the tested grid sizes, single precision was approximately 2.4–2.5 times faster than double precision. This is however specific to the hardware used; in the following precision comparison, it was a Tesla T4 GPU with a FP64 ratio of 1:32.

Table 3.1 | Runtime and speedup for single and double-precision LibTorch simulations on a Tesla T4 GPU. The speedup is the double-precision runtime divided by the single-precision runtime.

N	Double (s)	Float (s)	Speedup	Linear speedup	nonlinear speedup
2^{20}	9.4	3.9	2.4×	4.1×	2.0×
2^{21}	18.1	7.4	2.4×	4.1×	2.0×
2^{22}	36.3	14.8	2.5×	4.2×	2.0×

The precision comparison introduces a measurable accuracy cost, summarised in Table 3.2. The maximum field difference is measured relative to the peak of the double-precision field. After 1,000 steps the difference remains small, but it grows substantially after 10,000 steps. The float run also loses energy, with greater loss over the longer run.

Table 3.2 | Difference and energy change for single-precision LibTorch relative to double-precision LibTorch on a Tesla T4, using the symmetric split-step method with $N = 2^{20}$. The maximum field error is normalised by the peak of the double-precision field.

Steps	Maximum field error / peak	Energy change (float)
1,000	5.1×10^{-4}	−0.043%
10,000	5.2×10^{-2}	−0.44%

An additional dispersion-only test and a direct FFT round-trip test help identify the source of this energy loss. With nonlinear effects disabled, 1,000 single-precision steps produced an energy change of -4.3×10^{-4}. Applying 1,000 forward and inverse complex64 FFT round trips to a field of size 2^{20} produced a similar change (-4.25×10^{-4}), whereas complex128 changed the energy by approximately 4×10^{-13}. This indicates that the observed loss is associated with single-precision FFT round-off in cuFFT rather than the propagation phase itself. Repeated small downward changes can accumulate over many steps. The separate error-growth test at 8,192 points shows that the maximum field error grows approximately linearly with step count when Kerr is disabled, and approximately quadratically when Kerr is enabled over the measured range. The Kerr-enabled curve follows the plotted slope-two guide, while the dispersion-only curve grows more slowly. This supports the interpretation that nonlinear effects compound the numerical differences; these growth rates describe the tested configuration and range.

3.2.3 Summary

The measurements show that GPU acceleration is most useful once the workload is sufficiently large. With four channels and SPM/XPM enabled, the GPU also outperformed the CPU for the tested grid sizes. The timing breakdown shows that the linear and remaining parts of the simulation contribute differently to total runtime. Although using single precision gave an approximately 2.4-fold speedup over double precision for the tested grid sizes, the error in the results makes it unusable past generating a quick answer where exact precision isn't required. The exact results are specific to the tested hardware and configurations.

4 Discussion

4.1 Future Work

5 Conclusion

This work extended the simulation to support GPU-backed field data and custom pulse profiles. By adding custom defined functions, users are now able to define their own wave and spatially dependant fields without changing any code. The research then implemented and proved that GPU acceleration can greatly accelerate simulation, and therefore research speed. GPU acceleration was seen to have greater benefit as the workload got larger, through increasing grid-size, or number of pulses. For small grids, the time taken to launch GPU operations can make the GPU slower than the CPU. For larger grids and more channels, the GPU was substantially faster, reaching roughly 19x improvements.

The outcome of this research is that the SBS simulation is now capable of utilising the previously unrealised computational power of the GPU.

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A Appendix

TODO: Appendix

A.1 Cash–Karp RK5 Stages

$$
\begin{aligned}
A^{(i)} &= U^{\mathcal{D}}_{h/2} A_n, \\
k_1 &= U^{\mathcal{D}}_{h/2} f(A_n, z_n), \\
k_2 &= U^{\mathcal{D}}_{3h/10} f\Big(U^{\mathcal{D}}_{-3h/10}\left[A^{(i)} + \tfrac{h}{5}k_1\right],\ z_n + \tfrac{h}{5}\Big), \\
k_3 &= U^{\mathcal{D}}_{h/5} f\Big(U^{\mathcal{D}}_{-h/5}\left[A^{(i)} + h\left(\tfrac{3}{40}k_1 + \tfrac{9}{40}k_2\right)\right],\ z_n + \tfrac{3h}{10}\Big), \\
k_4 &= U^{\mathcal{D}}_{-h/10} f\Big(U^{\mathcal{D}}_{h/10}\left[A^{(i)} + h\left(\tfrac{3}{10}k_1 - \tfrac{9}{10}k_2 + \tfrac{6}{5}k_3\right)\right],\ z_n + \tfrac{3h}{5}\Big), \\
k_5 &= U^{\mathcal{D}}_{-h/2} f\Big(U^{\mathcal{D}}_{h/2}\left[A^{(i)} + h\left(-\tfrac{11}{54}k_1 + \tfrac{5}{2}k_2 - \tfrac{70}{27}k_3 + \tfrac{35}{27}k_4\right)\right],\ z_n + h\Big), \\
k_6 &= U^{\mathcal{D}}_{-3h/8} f\Big(U^{\mathcal{D}}_{3h/8}\left[A^{(i)} + h\left(\tfrac{1631}{55296}k_1 + \tfrac{175}{512}k_2 + \tfrac{575}{13824}k_3 + \tfrac{44275}{110592}k_4 + \tfrac{253}{4096}k_5\right)\right],\ z_n + \tfrac{7h}{8}\Big).
\end{aligned}
\tag{A.1}
$$

$$
A^{(5)}_{n+1} = U^{\mathcal{D}}_{h/2}\left[A^{(i)} + h\left(\tfrac{37}{378}k_1 + \tfrac{250}{621}k_3 + \tfrac{125}{594}k_4 + \tfrac{512}{1771}k_6\right)\right]. \tag{A.2}
$$